



DEALER BEST PRACTICE SERIES

Selective Cleaning Using Dry Ice

Maintenance and Repair Application Component Life Management Component Renewal (CRC) MARC Management

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1.0 Introduction

The importance of clean machines and components when performing maintenance and repair tasks has been understood and promoted for a long time. Traditional cleaning methods with high pressure water cannons and steam clean units will continue to play a vital role, though in certain circumstances and applications, it is difficult to achieve the desired cleanliness standards.

This document describes the trial and results of Cryogenic or Dry Ice Cleaning at an Australian coal mine. The focus was the engine and pump rooms of a Cat 5230 excavator. The mine operation intends to continue the use of this method for select cleaning jobs such as in-frame component overhauls when it is critical to have a clean working environment. The process is also extremely effective when cleaning electrical connectors and components as well as hydraulic and pneumatic hoses and connectors.

2.0 Best Practice Description

The trial of this cleaning process was conducted in the engine and pump rooms of a 5230 excavator. The purpose was to deliver a cleaner working environment for fitters. This allows for improved identification of mechanical and electrical faults. A local dry ice cleaning contractor performed the cleaning.

The work may be conducted in any safe location at the mine. The process does not require machine cool down (other than for any safety related reasons for those performing the cleaning). The process generates a moderate level of noise, so it is recommended that the cleaning operation be conducted within an area requiring hearing protection.

Cryogenic cleaning uses dry ice pellets (CO_2 in solid form) as the cleaning media. The dry ice pellets are directed at high speed at the surface to be cleaned. The intended surfaces are cleaned very effectively and quickly via a combination of 1) Kinetic energy, 2) Low temperature thermal energy, and 3) sublimation (instantaneous conversion of the pellets from a solid to vapor state).

The following illustrations describe the physical processes that take place during the cleaning operation.



- 1) Kinetic. When dry ice pellets are accelerated in a jet of compressed air and strike a surface at the speed of sound, they crack and loosen the coating on the surface being treated.

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- 2) Low temperature thermal. The low temperature of dry ice pellets (-79C) makes the coating brittle, cracks it and loosens it. This allows dry ice to permeate the coating.



- 3) Sublimation. Dry ice penetrates the coating and immediately sublimates (passes direct from solid to vapor state). This results in a 700– fold increase in volume, an explosive effect that lifts the coating off the surface.

The process uses compressed air and a variety of nozzle configurations to deliver and direct the dry ice. The velocity and distribution of the media can be controlled to remove only the desired contaminants and will not disturb paint, decals, labels, registration marks, etc. Since the dry ice pellets convert into a vaporous state, there is little to no residual material left (mainly dirt and grit that is cleaned from the machine or components). Chemicals or solvents are not used so contaminated water treatment is not required. Clean up and disposal time is minimal and drying time is eliminated.

There are a number of mine operations in the region currently using this method for select cleaning jobs. The process is extremely effective when used for in-frame component overhauls, major PCR work, cleaning of draglines, cleaning of electrical components and cabinets, or any application where it is critical to have a clean, dry working environment.

The following pictures were taken during the trial:

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Before

After



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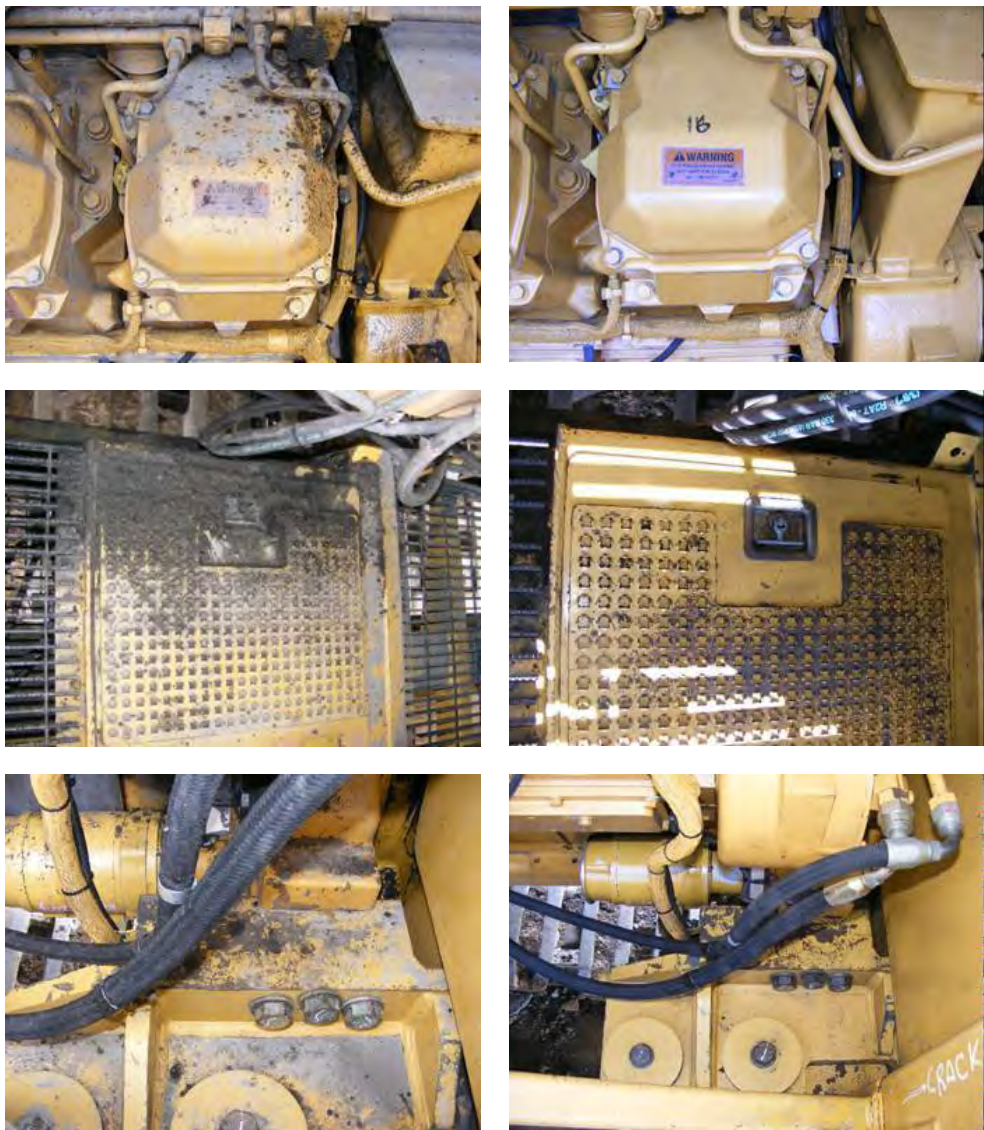
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Before

After



3.0 Implementation Steps

Few mine operations are likely to invest in dry ice cleaning facilities or equipment. Therefore, if there are dry ice cleaning contractors within the mine vicinity, implementing this process is as simple as engaging the services of local suppliers. The cleaning operations can be performed in any location allowed by the mine operation management. No special preparations are required.

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For the job described above, it took 1 hour to set up, two dry-ice units for 5 hours and another hour to pack up. Based upon this:

Set up/Pack up:	AUD\$150.00 per hour x 2 = AUD\$ 300.00
Labor:	AUD\$240 per hour x 5 = AUD\$1,200
Air:	AUD\$ 130.00 per hour x 5 = AUD\$650
Dry Ice Consumption:	AUD\$300.00 per hour x 5 = \$ 1,500
Travel:	May be applicable
Total:	AUD\$3,650

(These results are for comparative purposes. Actual costs may differ depending on location.)

4.0 Benefits

The list of benefits is significant. Here are a few of the major advantages delivered and realized as a result of the trial:

- Fast and efficient
 - During the above trial the equipment was completely decontaminated in 5 hours. It was estimated that it would have taken at least 10 hours with normal 'wet' methods to achieve a significantly lesser result.
- Many time consuming and labor intensive steps are no longer required, including:
 - Cool down, disassembly, transport of equipment and reassembly.
- Improve electrical equipment reliability
 - As with so many sites, electrical issues due to damaged harnesses and connectors are a major cause of machine stoppages. The local dealer recognizes that introducing high-pressure water into harnesses and connectors creates many of these problems. Further, in many cases, the mine water used for cleaning is highly saline so the development of corrosion on electrical contacts develops over time. Dry Ice cleaning removes the possibility of water being introduced into critical areas.
- A very clean result. The photos above show how dramatic the results are, even in hard to reach areas. Technicians can easily identify faults and defects on thoroughly cleaned components.
- High resistance paths in electrical connections caused by build up of dust and dirt over time is believed to be the cause of many electrical faults. This dust and dirt can be effectively cleaned using dry ice.
- Due to the sublimation of the dry ice (turning to vapor) there are no secondary wastes left behind. This reduces costs for disposal, containment and collection of wastes and significantly reduces clean-up time.
- Will not damage hydraulic/pneumatic hoses, electrical cable insulation or electronic components.
- No chemicals required.

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5.0 Resources Required

A dry ice cleaning contractor in the mine operation area may conduct the cleaning.

Purchase of the dry ice cleaning equipment and operation training may also be an option.

6.0 Supporting Attachments / References

Dry Ice Solutions – Presentation Dec 08.ppt (attached to this document)

7.0 Related Best Practices

None at this time.

8.0 Acknowledgements

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Special mention must be made of the following two organizations for their willingness to share this information:

Hastings Deering Australia
CQ Dry Ice Solutions (www.cqdis.com.au)

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